Siefeldin Sobih

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I enjoy solving problems with math and ML. I have 3+ years of research experience in data science, building statistical models across diverse datasets. My current interests are interpretable AI and LLM integration in variety of problems.

RESEARCH EXPERIENCE

Utah State University - Department of Mathematics - Kevin Moon Lab 01/2026 – Present
Graduate Research Assistant Logan, UT

- Working to integrate vision-language models and large language models into robot-to-robot communication so robots can share and interpret observations without relying on a central controller.
- Adapting algorithms that merge object annotations across robots' maps so observations of the same object can be linked without creating duplicate entries.

Utah State University - Department of Biology - Karen Kapheim Lab 08/2023 – 12/2025
Graduate Research Assistant Logan, UT

- Analyzed gut microbiome and health data in R using mixed-effects models and multivariate tests to assess relationships between health measurements and microbiome composition.
- Developed a regression model to use glucose measurements from two different instruments in the same analysis, predicting Accu-Chek readings from i-STAT measurements with a cross-validated mean absolute error of 9.0 mg/dL (RMSE: 12.8 mg/dL).
- Automated data-analysis workflows with Nextflow and adapted them to run on a high-performance computing cluster.
- The project led to two pending publications that expand our knowledge on two endangered subspecies.

Nile University - Department of Bioinformatics 01/2021 – 05/2021
Intern Research Assistant Giza, Egypt

- Developed the branching logic for a Python bioinformatics pipeline that checks dependencies, prompts users to select a tool, passes their inputs through Bash commands to run it, and collects the output. Our work is currently published
- Developed a Python script to retrieve PAM250 scores for aligned amino acid pairs between humans and other species, replacing repetitive manual matrix lookups for our intern cancer-research team.

EDUCATION

Utah State University - Department of Mathematics 08/2026 – Present
PhD in Statistics and Data Science Logan, UT

Utah State University - Department of Biology 08/2023 – 12/2025
MS in Computational Biology Logan, UT

Cairo University - Faculty of Agriculture - Biotechnology program 08/2017 – 05/2021
BS in Agricultural Biotechnology Giza, Egypt

HACKATHONS

Lucid Hackathon 11/2026
Built a rule-based bot for Lucid's 2026 real-time strategy competition and advanced to the semifinals. Logan, UT

IBM July's creative builders challenge 07/2026
Led a team in building VNForge, a Python desktop app that turns written scenes into editable dialogue, player choices, and branching storylines through AI APIs. Virtual, USA

TECHNICAL SKILLS

AI/ML — LLM integration, evaluation and deployment; PyTorch; vLLM; GLMs; LMMs; ANOVA variants.
Engineering — Git; Conda; Nextflow; Python; Linux.